University of London

ST3189 MACHINE LEARNING COURSEWORK

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1. INTRODUCTION

Machine Learning is a set of computational techniques that enable systems to learn and identify patterns and relationships within data to continuously improve predictions, refine performance, and enhance accuracy over time. These computational techniques are often used in business analytics to assist in data-driven decisions and operations. This study applies machine learning methods to a customer personality dataset to analyse various demographic characteristics, identify purchasing patterns across product categories, and gather responses to product marketing. This will provide an insight on customer behaviour for future marketing decisions. Unsupervised learning techniques are applied to identify customer characteristics, regression models are implemented to analyse consumer spending habits, and classification models are used to predict customer responses to marketing initiatives.

2. DESCRIPTION OF DATASET

The dataset being trained for this study is a Customer Personality Analysis dataset, containing 2,240 observations and 29 variables. The variables include characteristics such as year of birth, income level, marital status, as well as behavioural variables such as purchasing behaviour across product categories, store and web purchases, and responses to marketing campaigns. Table 1 displays a summary of key variables for this study. The aim of training this dataset is to better analyse consumer purchasing patterns to improve marketing strategies and cater to specific customer tastes and preferences. An initial analysis of the data provides an overview of the dataset features. Figure 1 displays a right skewed income distribution, with most customers falling in the low to middle income ranges of $20,000 to $80,000 and fewer high-income observations of $100,000 and above. Figure 2 shows the average spending across different product categories, with wine and meat products garnering the highest average expenditure. Figure 3 presents the variation of responses to past marketing campaigns, with majority customers not accepting any offers from it. This initial analysis introduces a useful insight to the data and the further analysis conducted by the machine learning methods.

	Variable	Description
1	Year_Birth	Year of birth of the customer
2	Education	Customer's education level
3	Marital_Status	Customer's marital status
4	Income	Annual household income
5	Recency	Days since the customer's last purchase
6	MntWines	Amount spent on wine products
7	MntMeatProducts	Amount spent on meat products
8	NumWebPurchases	Number of purchases made through the website
9	NumStorePurchases	Number of purchases made in physical stores
10	Response	0 = No (not accepted), 1 = Yes (accepted)

Table 1: Summary of key variables

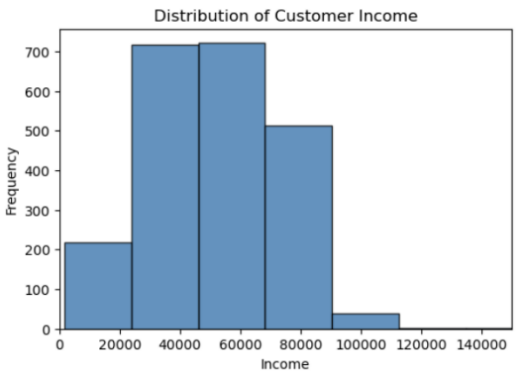


Figure 1.1: Distribution of Customer Income

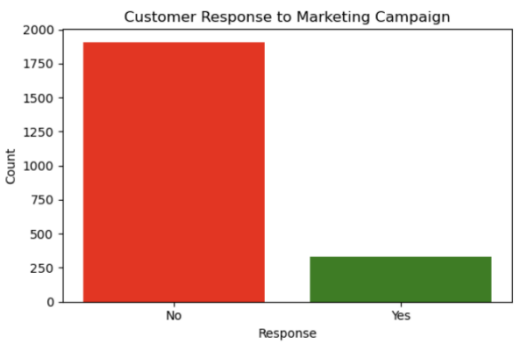


Figure 1.3: Customer Responses

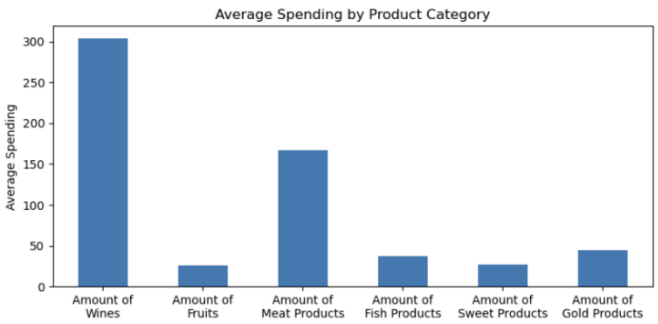


Figure 1.2: Average spending by Category

3. *UNSUPERVISED LEARNING*

3.1 AIM

Unsupervised learning methods will be applied to categorise customers with similar characteristics into clusters based on demographics and spending habits. By identifying groupings and relationships in the dataset, this analysis aims to create specific client profiles and uncover spending trends to enhance the effectiveness of marketing schemes and better cater to clientele.

3.2 RESEARCH QUESTIONS

The research questions (**RQs**) to examine:
RQ1: What are the distinct customer clusters based on demographics and spending habits?
RQ2: Which unique characteristics differentiate the customer clusters with regards to spending trends and client engagement?
RQ3: How do the distinct customer clusters enhance the effectiveness of marketing strategies?

3.3 VARIABLES

Numerical variables related to demographics and spending habits will be selected, while categorical variables such as education and marital status are left out to avoid unnecessary complications with distance-based approaches. The Year_Birth variable will be transformed into Age to improve interpretation, and Total_Spending for expenditure across product categories and Total_Purchases for combined channel purchases will be created as aggregated features to provide a useful overview. Missing and duplicate values are dropped to prevent any bias, and the variables are standardised with z-score normalisation to ensure equal importance in clustering. The final variables used will be Age, Income, Recency, Total_Spending, and Total_Purchases.

3.4 METHODOLOGY

Clustering techniques such as K-means and Hierarchical clustering, and dimensionality reduction techniques such as Principal Component Analysis (PCA) will be employed to better organise the data. This will simplify the analysis, categorisation and visualisation to improve comparability among variables (Hastie, Tibshirani and Friedman, 2009). These methods applied on standardised variables will further improve consistency and accuracy.

3.5 ANALYSIS

K-means clustering is applied to group data points with similar characteristics into k number of clusters by calculating the minimum Within-Cluster Sum of Squares (WCSS). The algorithm iteratively assigns each point to the nearest cluster using Euclidean distance and repeats this until WCSS does not change against the number of clusters formed. Figure 2.1 is an Elbow plot with WCSS plotted against number of clusters. The decrease in WCSS is rapid from k=1 to k=3 and then slows, creating an 'elbow' from k=3 onwards. This indicates that WCSS changes marginally after 3 clusters, suggesting that 3 clusters are optimal. Figure 2.2 is a Silhouette plot with silhouette score plotted against number of clusters. The k cluster with a higher score indicates optimal clustering. The highest score is at k=2, however the score remains reasonably high at k=3, indicating strong segmentation with 3 clusters and supports the analysis of 3 optimal clusters.

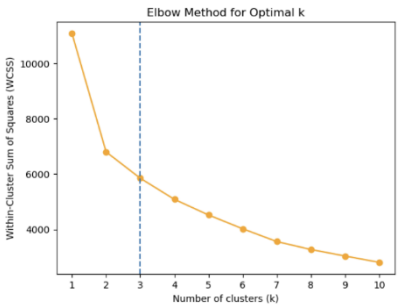


Figure 2.1: Elbow Plot

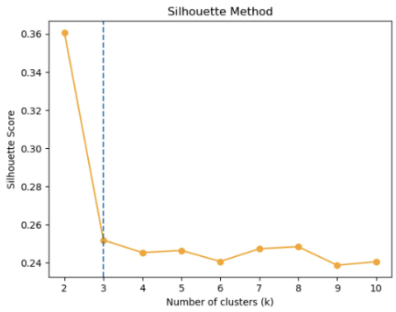


Figure 2.2: Silhouette Plot

Cluster	Count	Age	Income	Recency	Total_Spending	Total_Purchases
1	586	52.22	36436.79	23.98	152.38	7.14
2	876	57.57	71858.29	48.46	1196.53	19.43
3	572	54.61	38804.07	74.92	172.17	7.56

Table 2.1: Cluster Profiles

Table 2.1 above presents the cluster profiles and sizes determined with k=3 clustering. The results indicate differences in income, spending and purchases. Cluster 1 (586 customers) represents the 'lower-value' customers with lowest average income, recency rate, spending and purchasing activity. Cluster 2 (876 customers) represents 'high-value' customers and is the largest. It has highest average income, spending, and purchases. A moderate recency rate indicates higher value purchases when customers engage. Cluster 3 (572 customers) represents 'ordinary' customers, with lower average income and moderate spending and purchasing activity, although a high recency rate indicates stable engagement of purchases. This indicates a group with less purchasing power and expendable incomes. Overall, the 3 clusters are segmented by income, spending, and purchasing, which depicts the value they contribute to revenue.

Hierarchical clustering is applied to support the clusters determined using K-means. Ward's linkage is used to iteratively combine similar points by minimising within-cluster variance. A dendrogram is generated to visualise the optimal number of clusters and illustrate the cluster structure. Figure 2.3 shows the dendrogram following Ward's method.

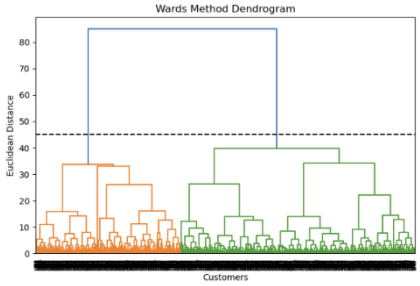


Figure 2.3: Dendrogram

The clear separation of main branches and a black line generated at an approximate distance of 45 shows 3 segmented clusters. A large vertical distance between merges implies that grouping clusters beyond that will likely distort similarities. This supports the above K-means clustering results of 3 optimal clusters and validates a distinct cluster structure apparent in this dataset.

Principal Component Analysis (PCA) is a dimensionality reduction technique applied to simplify high-dimensional data in the dataset into a smaller set of variables, and maximise key information preserved. This allows a clearer visualisation of underlying clusters, patterns, and structures. Table 2.2 and Figure 2.4 below show outputs of the PCA.

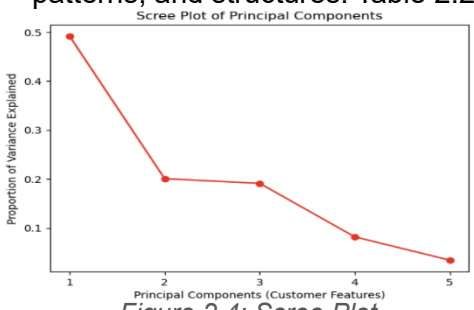


Figure 2.4: Scree Plot

	Principal Component	Explained Variance
1	PC1	0.488228
2	PC2	0.200995
3	PC3	0.190397
4	PC4	0.085350
5	PC5	0.035030

Table 2.2: Variances

	PC1	PC2	PC3	PC4	PC5
Age	0.176286	0.290377	0.938354	-0.042289	0.048051
Income	0.533586	-0.038047	-0.045878	0.838404	-0.093875
Recency	0.005201	0.955162	-0.294953	0.022620	-0.011392
Total_Spending	0.587919	-0.033969	-0.151143	-0.301756	0.734368
Total_Purchases	0.581836	-0.027301	-0.086870	-0.451356	-0.670412

Table 2.3: Components Loadings

PC1 and PC2 account for around 48.82% and 20.09% of total variance respectively. This shows that just 2 PCs explain approximately 70% of total variance.

Figure 2.4 shows a Scree plot with a slow decline in explained variance after the first 2 components, implying that extra components do not add benefits and that fewer components are enough to retain maximum information. The loadings from Table 2.3 show that **PC1** is significantly influenced by Spending, Purchases, and Income, thus representing '**Customer Value**'. **PC2** is highly influenced by mostly Recency and Age, thus representing '**Customer Activity**'. Overall, these findings reinforce that fewer components can still efficiently represent data points and retain key behaviours. Thus, this supports the clusters formed above.

3.6 RESULTS

RQ1: K-means and Hierarchical clustering identify 3 optimal distinct customer clusters: High-value customers, Ordinary Customers, Lower-value customers.

RQ2: The clusters are differentiated mostly by characteristics such as income, spending, and purchases: High-value customers have high income, spending, purchases, Ordinary customers have average behaviour, Lower-value customers have low spending with low engagement.

RQ3: Targeted marketing would work best: Personalised offers for High-value customers, subscriptions for Ordinary customers, and discounts or promotions for lower-value customers.

4. REGRESSION

4.1 AIM

Regression methods will be applied to model and predict customer spending behaviour. By investigating the relationship between demographics and spending and behavioural variables, this analysis aims to establish key determinants of customer spending to enhance marketing decisions.

4.2 RESEARCH QUESTIONS

The research questions (**RQs**) to examine:
RQ1: Which factors have the highest influence on customer total spending?
RQ2: How accurate are the regression models in predicting customer total spending?
RQ3: Which regression model has the most accurate predictive performance?

4.3 VARIABLES

Customer demographics and behavioural variables will be used as predictors of total spending in the regression models. The dependent variable is Total_Spending based on overall expenditure, while independent variables are demographic and behavioural variables such as Age, Income, Recency, and Total_Purchases. Other variables contributing to total spending will be excluded to avoid multicollinearity and target leakage. Duplicate and null values will also be excluded.

4.4 METHODOLOGY

Regression methods such as Linear Regression will be employed to model linear relationships, while Random Forest (Breiman, 2001) and Gradient Boosting (Friedman, 2001) will model more dimensional, non-linear relationships. These will show predictions of expenditure and comparisons of model performance.

4.5 ANALYSIS

Linear Regression assumes a linear relationship between predictors and the target variable; hence it is applied to determine the relationship between customer demographics and total spending. The algorithm calculated an R^2=0.699, as seen in Table 3.1, implying that around 70% of variance in customer spending is accounted by the predictors. Figure 3.1 is a scatter plot illustrating a diagonal relationship between actual and predicted spending, with a clear deviation as spending increases.

Model	R^2	MAE	RMSE
Linear Regression	0.699097	238.27099	337.934029

Table 3.1: Linear Regression Model Performance

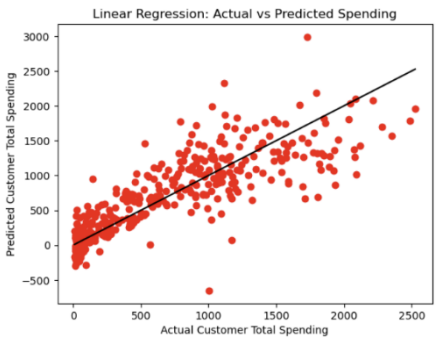


Figure 3.1: LR Scatter Plot: Actual vs Predicted

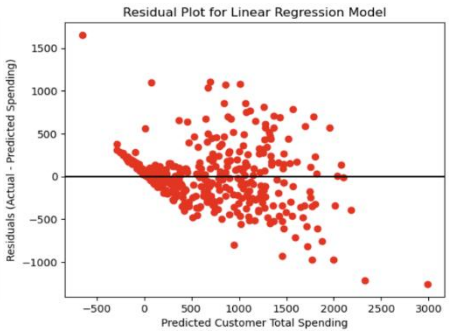


Figure 3.2: Residual Plot

Figure 3.2 is a residual plot also showing increased variance at higher spending levels. These models indicate the presence of heteroscedasticity, implying that linear regression may not be appropriate to visualise accurate customer spending behaviour.

Random Forest is an ensemble learning method that builds decision trees and combines their predictions to obtain a more reliable result. It is applied to model complex non-linear relationships between variables, which produces a more accurate model performance as

Model	R^2	MAE	RMSE
Random Forest	0.867669	120.327076	224.103494

Table 3.2: Random Forest Model Performance

compared to linear relationships. The algorithm calculated an R^2=0.868, as seen in Table 3.2, showing a significant enhancement in analytic performance as now around 87% of variance in customer spending is accounted by predictor variables. Figure 3.3 is a scatter plot similarly illustrating a diagonal trend between actual and predicted spending, however the predictions are closer and there is lesser deviation as spending increases. Figure 3.4 is a feature importance plot which displays that NumCatalogPurchases has the highest significance as a variable, followed by Income and Total_Purchases. This suggests that expendable income and purchasing power are the key determinants of spending. Compared to Linear Regression, predictions deviate lesser from actual values, indicating a better model performance.

Figure 3.3: RF Scatter Plot: Actual vs Predicted

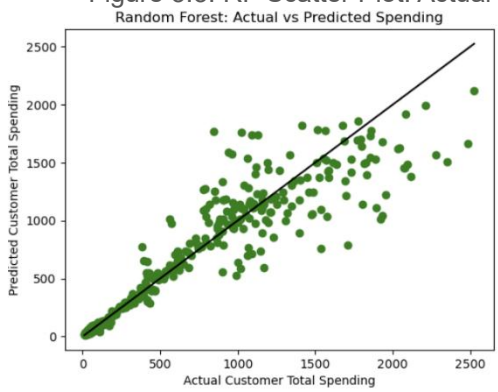
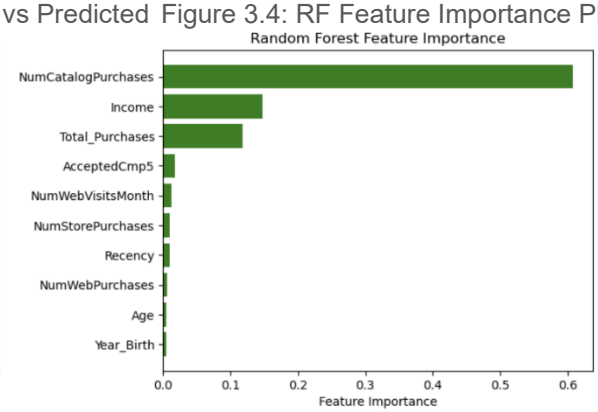


Figure 3.4: RF Feature Importance Plot



Gradient Boosting is another ensemble learning method applied to builds models one by one, with each new model fixing the residual errors of the total ensemble and sequentially improving prediction performance. The algorithm calculated an R^2=0.855, as seen in Table 3.3, suggesting that around 85% of variance in customer spending is accounted by the predictors. While this signifies a strong predictive model, it is still marginally lower than Random Forest. Figure 3.5 is the scatter plot of actual against predicted spending for Gradient Boosting, presenting a similar trend and low deviation.

Model	R^2	MAE	RMSE
Gradient Boosting	0.855911	139.267156	233.847981

Table 3.3: Gradient Boosting Model Performance

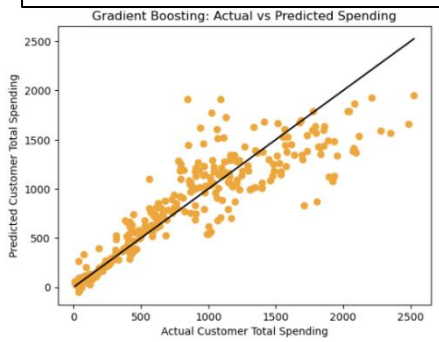


Figure 3.5: GB Scatter Plot: Actual vs Predicted

Figure 3.6: GB Feature Importance Plot

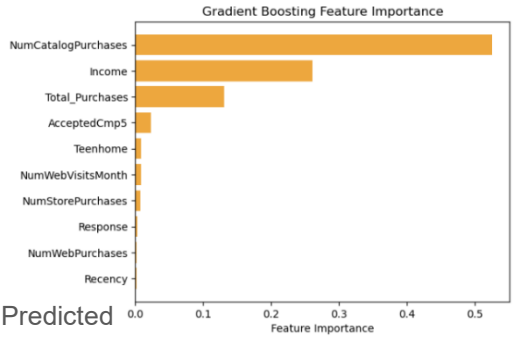


Figure 3.6 is a feature importance plot similarly presenting NumCatalogPurchases as the highest influence, with Income and Total_Purchases following. Overall, these models reiterate the significance of consumer behaviour and spending power.

4.6 RESULTS

RQ1: The behavioural factors NumCatalogPurchases, Income, and Total_Purchases have the highest influence on customer total spending.

RQ2: The regression models show strong predictive performance, with Random Forest and Gradient Boosting holding higher accuracy than Linear Regression.

RQ3: The Random Forest holds strongest accuracy, with the highest R^2 value and lowest error metrics among all the models, as seen in Table 3.4.

Model	R^2	MAE	RMSE
Linear Regression	0.699097	238.270990	337.934029
Random Forest	0.867669	120.327076	224.103494
Gradient Boosting	0.855911	139.267156	233.847981

Table 3.4: Comparison Table of Regression Model Performance Metrics

5. CLASSIFICATION

5.1 AIM

Classification techniques wil be applied to project customer responses to marketing strategies. By investgating the pattern between the population and behavioural factors and customer responses, this analyis will determine the siginificant variables that affect marketing strategies.

5.2 RESEARCH QUESTIONS

The research questions (**RQs**) to examine:

RQ1: What are the most influential factors of customer responses to marketing strategies?

RQ2: How accurate are the classification models in predicting customer responses?

RQ3: Which classification model has the most accurate predictive performance?

5.3 VARIABLES

Several variables related to customer responses will be selected. The dependant variable is a binary variable Response, which generates 1 if the customer accepted the strategy, and 0 if not. The independent variables consist of influential behavioural attributes and characteristics, such as Age, Income, Recency, and Total_Purchases. This range will provide an overview into purchasing power and habits. Other variables relevant to marketing interactions will be included, while direct contributors to response outcomes, null, and duplicates will be excluded to prevent target leakage.

5.4 METHODOLOGY

Classification techniques such as Logistic Regression will be applied as a foundation to showcase binary classification, while Random Forest and Gradient Boosting will model more intricate non-linear patterns. The predictive power of accurate classification by these models will be compared by Accuracy, Precision, Recall, F1-scores, and ROC-AUC.

5.5 ANALYSIS

Logistic Regression is employed as the foundational model of customer response prediction. The logistic regression model attained an accuracy score of 0.838 and ROC-AUC of 0.787, as seen in Table 4.1, suggesting strong predictive power. The confusion matrix in Figure 4.1 shows that the model correctly identifies a majority of 311 non-responders but is unable to correctly place the responders, with only 30 correct and 36 misclassified, thus producing a low recall score of only 0.455. Figure 4.2 illustrates the ROC curve curving away from the baseline, indicating that this model achieves higher accuracy than ordinary classification. These indicators imply that Logistic Regression is a good base for classification, however its poor recall indicates limited capability in categorising responders and that complex models may be better suited for more descriptive classification.

Figure 4.1: Logistic Regression Confusion Matrix

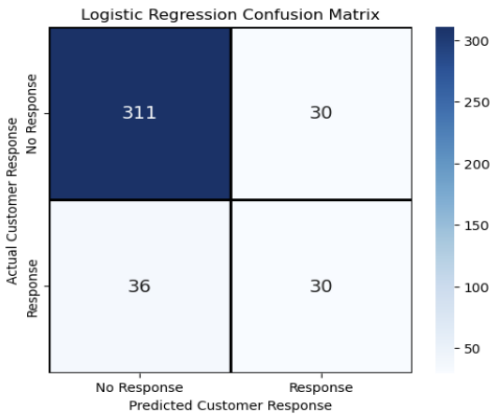
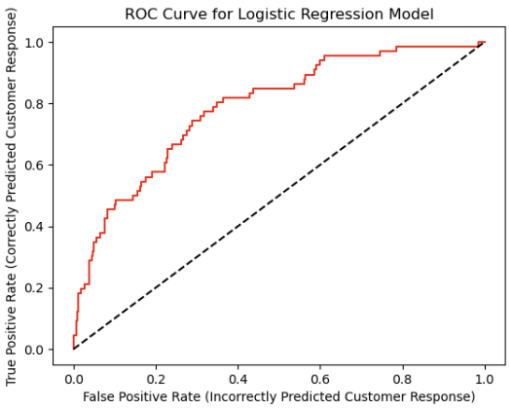


Figure 4.2: Logistic Regression ROC Curve



Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.838	0.5	0.455	0.476	0.787

Table 4.1: Logistic Regression Model Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.855	0.706	0.182	0.289	0.873

Table 4.2: Random Forest Model Performance

Random Forest is an ensemble method employed again to present more complex non-linear interactions and patterns in the dataset. The random forest model attained an accuracy score of 0.855 and an ROC-AUC of 0.873, as seen in Table 4.2 above, signifying an improvement in predictive power as compared to the logistic regression model. The confusion matrix in Figure 4.3 shows that this model correctly identifies a majority of 336 non-responders, but identifies few responders, with only 12 correct and 54 misclassified, thus producing a low recall score of 0.182. The ROC Curve in Figure 4.4 presents a larger ROC-AUC with the curve clearly further away from the baseline, reflecting a strong accuracy model.

Figure 4.3: Random Forest Confusion Matrix

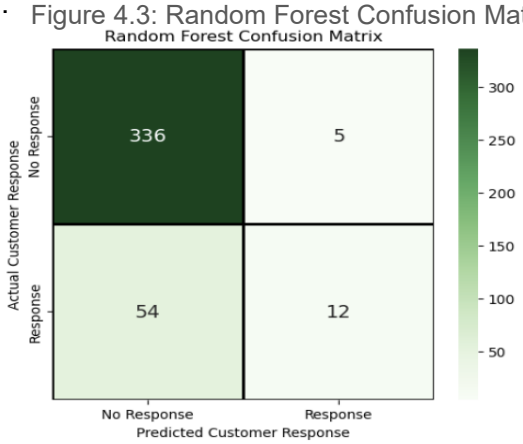
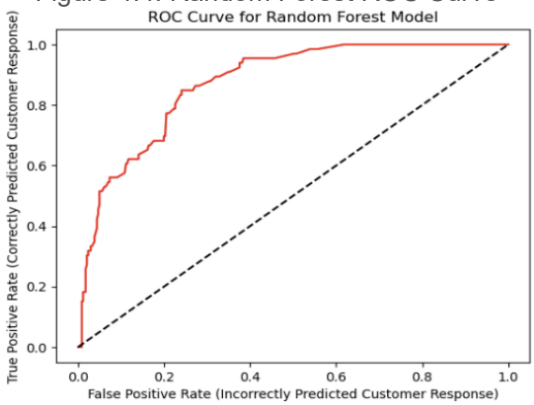


Figure 4.4: Random Forest ROC Curve



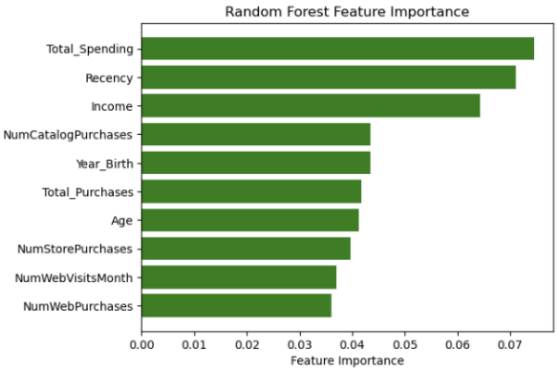


Figure 4.5: Random Forest Feature Importance Plot

A feature importance plot is included, as seen in Figure 4.5, to illustrate key drivers, with Total_Spending, Recency, and Income being the key drivers and signifying that spending habits and customer interactions are influential predictors of customer responses to marketing strategies. As compared to Logistic Regression, the Random Forest model shows an improved predictive power and discriminative ability, however its recall is lower. These indicate a compromise between accuracy and precision, leading to a similar limited capability in classification.

Table 4.3: Gradient Boosting Model Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Gradient Boosting	0.877	0.722	0.394	0.51	0.903

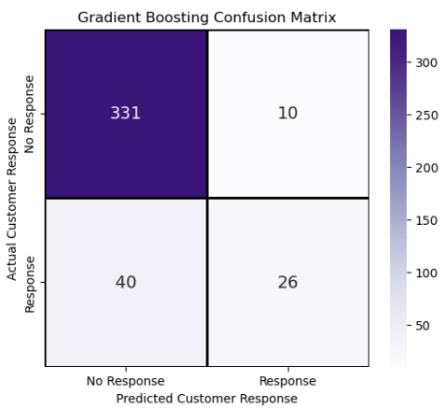


Figure 4.6: Gradient Boosting Confusion Matrix

Gradient Boosting is another ensemble learning method applied to classification to create models one by one, with each new one fixing the residual errors of the total ensemble and sequentially improving prediction performance. The gradient boosting model attained an accuracy score of 0.877 and ROC-AUC of 0.903, as seen in Table 4.3 above, showing the strongest overall performance among the 3 models. The confusion matrix in Figure 4.6 shows 331 correctly identified non-responders, and an increase in identified responders, with 26 correct and 40 misclassified, thus producing a higher recall of 0.394 as compared to Random Forest.

Figure 4.7: Gradient Boosting ROC Curve

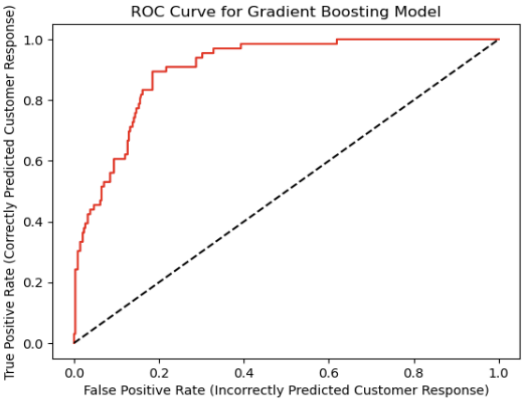


Figure 4.8: Gradient Boosting Feature Importance Plot

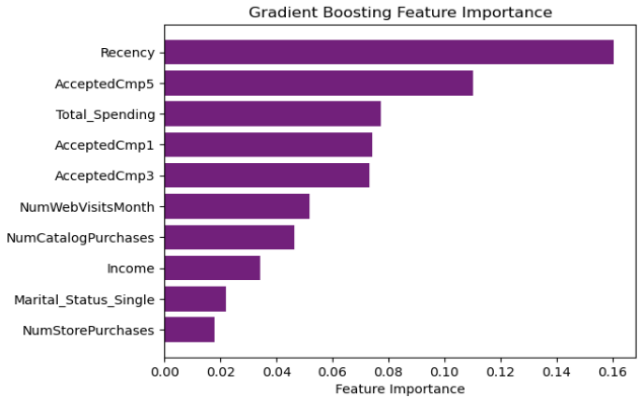


Figure 4.7 presents the ROC curve nearing the top-left corner, accounting for the highest ROC-AUC and signifying the strongest discrimination. The feature importance plot in Figure 4.8 displays Recency, AcceptedCmp5 being past responses, and Total_Spending as the key drivers, signifying that predictions of responses to marketing strategies are highly dependent on customer engagement and previous involvement with any marketing campaigns. In comparison to the other models, Gradient Boosting projects the strongest overall performance, showcasing its competence in classification.

5.5 RESULTS

RQ1: The most influential factors of customer responses to marketing strategies are Recency, AcceptedCmp5 (past responses), and Total_Spending.
RQ2: The classification models show strong predictive power, with Gradient Boosting and Random Forest holding higher overall accuracy over and Logistic Regression.
RQ3: Gradient Boosting projects the most accurate predictive performance by generating the highest ROC-AUC, Accuracy, Precision, and F1 score, as seen in Table 4.4, which displays the strongest balance.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.838	0.500	0.455	0.476	0.787
Random Forest	0.855	0.706	0.182	0.289	0.873
Gradient Boosting	0.877	0.722	0.394	0.510	0.903

Table 4.4: Comparison Table of Classification Model Performance Metrics

6. CONCLUSION

The aim of this study was to analyse a customer personality analysis dataset to gain insights into customer behaviour to support data-driven marketing decisions. Multiple machine learning techniques were applied to gather information and study patterns in the data.

Unsupervised learning methods uncovered the various customer clusters grouped by similar demographics and spending habits, which created specific client profiles by certain spending trends. This would assist in generating marketing initiatives better catered to clientele interests, such as personalised offers for high spenders, subscriptions and loyalty programs for ordinary spenders, and promotions, discounts, and sales for low spenders. Regression techniques studied characteristics, spending, and behavioural habits to identify the key drivers of spending. The influential factors identified as income and overall purchasing behaviour are beneficial in understanding that high-value and frequent spenders should be prioritised in marketing schemes. Classification methods predicted customer responses to marketing by examining behavioural aspects and past responses. The key determinants established as recency, past responses, and total spending highlight the significance of optimising responses based on engagement and responsiveness. Overall, these machine learning methods provided insights that businesses could leverage to better target specific customer segments, prioritise important customers, and optimise marketing campaigns.

The effectiveness of the models in differed across the machine learning. In unsupervised learning, K-means clustering was applied as the primary method of obtaining the optimal

number of clusters and segmenting customers. Hierarchical clustering was support for validation and PCA provided cluster visualisation. In regression, Random Forest held the highest overall accuracy, indicating the importance of analysing non-linear relationships in the data. Lastly in classification, Gradient Boosting had the best predictive performance, indicating its high accuracy and balance of predictive metrics. Overall, this analysis showed that ensemble learning methods were most effective in uncovering underlying complex patterns in the dataset.

In conclusion, this study successfully showcased the application of machine learning techniques in the customer personality analysis dataset. With the combination of segmentation, prediction, and classification, useful business insights were attained for better targeting and marketing strategies.

7. REFERENCES

7.1 ACADEMIC REFERENCES

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7.2 DATASET LINK

The link to the Customer Personality Analysis Dataset is found below:
https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis